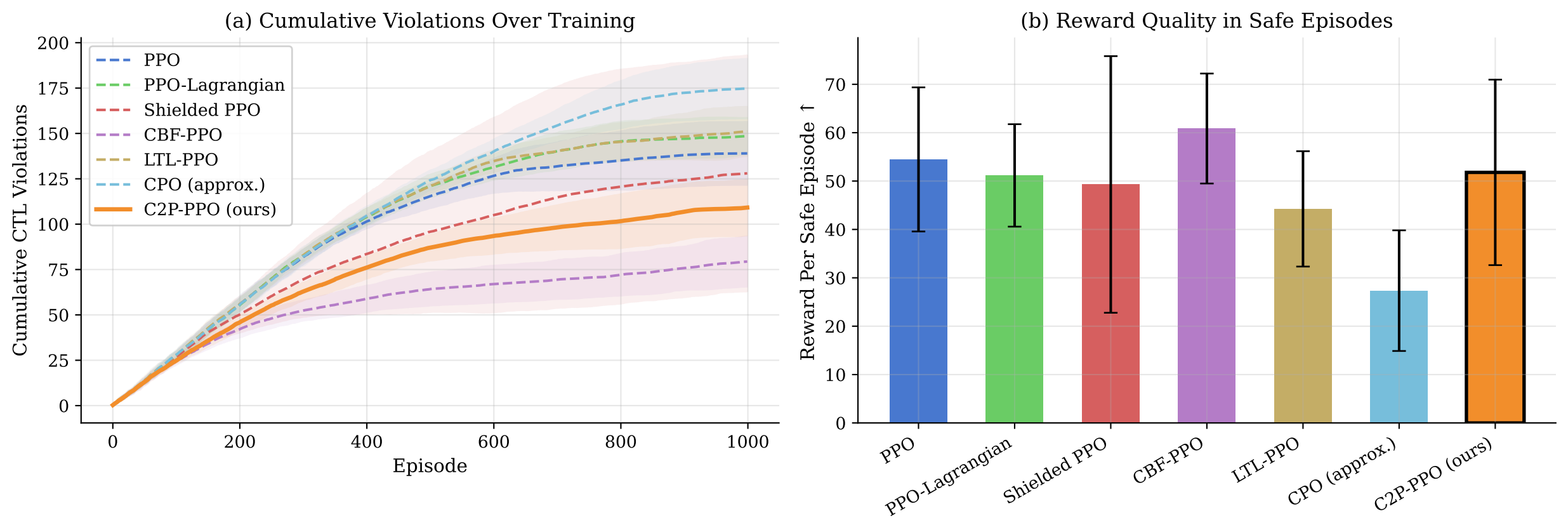

(a) Cumulative Violations Over Training
Cumulative CTL Violations
Episode
PPO
PPO-Lagrangian
Shielded PPO
CBF-PPO
LTL-PPO
CPO (approx.)
C2P-PPO (ours)
(b) Reward Quality in Safe Episodes
Reward Per Safe Episode ↑
PPO
PPO-Lagrangian
Shielded PPO
CBF-PPO
LTL-PPO
CPO (approx.)
C2P-PPO (ours)